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ITEC403

GRADUATION PROJECT ORIENTATION REPORT

<DORMITORY MANAGEMENT SYSTEM >

<GROUP NO. 5>

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CHAPTER 1

PROBLEM DEFINITION

Among this rapid advancement of today's technology, there's been a significant shift to digital platforms in practically every aspect of our daily lives. This shift is observed through the changes in the ways of communication, shopping, working, and even socializing where all these aspects have been integrated into online platforms with the key purpose of making individuals' lives more convenient. Nonetheless, despite these comprehensive changes and the integration of technology into every part of our lives, there is an evident oversight in particular areas, most prominently in the existence of a management system for student accommodations.

The project aims to tackle the problem originating from the lack of a unified online system for dormitory access and management within our institution. Students currently find themselves navigating through a mixture of dormitory websites, relying on word-of-mouth, or undertaking tedious manual searches. Navigating through numerous online sources students often encounter sites that remain outdated, missing vital academic year details, or platforms that are not fully equipped with the critical information and tools to aid students in their search. Additionally, while the reliance on acquaintances regarding dormitories can offer valuable insight, they are essentially subjective and may not accommodate to each individuals' specific wants and preferences. Those who pursue manual searches face their own struggles where they are obligated to physically visit various dormitories which can be fatiguing and time-consuming. Such unreliable methods not only consume excessive time but also heighten the potential for misinformation, making the hunt for the perfect living accommodation exceedingly challenging for students.

Such implications concerning dormitory access and management are more than just nuisances. Living conditions have such a significant influence on a student's well-being where a secure, pleasing, and supportive environment can contribute to a student's mental, physical, and academic health. Additionally, the reputation of the institution is highly influenced by level of easiness of locating and acquiring a suitable accommodation for the student. The establishment of

this unified system speeds up the process of attaining a housing therefore, aiding students in conserving their time, energy, and focus on their academic and personal ambitions.

CHAPTER 2

SCOPE OF THE STUDY

In the realm of education, where the happiness and ease of students is of highest importance the Dormitory Management System project strives to establish a developing and uplifting atmosphere, for students living in dormitories. Our method, for achieving the goals of this project is not just relevant but remarkably efficient and we are excited to share our vision.

2.1 Relevance And Effectiveness

Our project holds significance in the realm of education. With an increasing number of students choosing to reside on campus it has become important to have a management system that's efficient and caters to the needs of students. The implementation of the Dormitory Management System aims to improve the experience with and within dormitories. Both students and administrators will benefit from an efficient process.

2.2 Problem Management

We prioritize problem management as one of our areas of focus. We recognize that, in dormitory environment challenges are inevitable to occur. This project aims not to identify challenges but also to offer practical solutions. Our goal is to establish a system that enables resolution of issues finally improving the dormitory experience, for everyone involved.

2.3 Project Strategies

We have developed an approach that covers aspects, including student records management, room assignments and issue tracking. Our goal is to provide communication channels, a user interface, and an organized system, for keeping records. The strategies we implement aim to enable both students and administrators ensuring a trouble-free experience, in the dormitory.

2.4 Addressing Challenges

The Dormitory Management System project will address various challenges by providing real-time notifications for maintenance, ensuring transparent communication, and allowing room preference customization. We will utilize robust databases and user-friendly interfaces to simplify procedures.

To wrap things up, Dormitory Management System project not only resolves problems but develops the lives of students in dormitories. Our approach is applicable, efficient, and designed to bring a positive change. This project exemplifies our dedication to the benefit of students in the educational society.

CHAPTER 3

ADDITIONAL FEATURES

3.1 Booking and Reservation System

For the implementation of the booking and reservation system the following steps will be taken:

1. Design a database that can store information, about dormitories, rooms, availability status and bookings.
2. Develop an API that can handle booking requests and manage availability accordingly.
3. Build a user frontend interface where users can browse through dormitories check room availability and make their bookings.
4. Connect the API with the frontend interface to ensure integration.

For real time availability checks we can combine database queries with caching techniques. For instance maintaining a cache of rooms for each dormitory and updating it whenever there is a booking or cancellation. For payment processing we can integrate with established third-party payment processors like Stripe or PayPal. As for sending confirmation emails to users we can utilize mailing services such as SendGrid or Mailgun.

3.2 User Profile Customization

To enable users to create and customize their profiles as, per their preferences we will need to follow these steps:

1. Design a database that stores user profile details including name, email address, preferences and profile picture.

2. Develop an API that allows users to easily manage their profile information.

These steps should help us in implementing the desired features while ensuring user experience throughout the system. To enable users to access and modify their profile information we need to create a user interface (UI) on the frontend.

3.3 Notifications and Alerts

1. We must select a method of delivering notifications, such, as email, SMS or push notifications.
2. A backend API needs to be developed in order to send these notifications to users.
3. It is essential to integrate the API with the chosen notification delivery channel.

For instance when a user reserves a dormitory room it would be beneficial to send them a confirmation email. Additionally if they receive a message, from another user it would be helpful to notify them through chat.

3.4 Dormitory Owner Dashboard

1. Designing the dashboard to provide insights on occupancy rates, revenue and reviews.
2. Developing an API that retrieves data from the database.
3. Integrating the API with the dashboards user interface.

3.5 Review And Rating System

1. Adding fields to the database for storing review and rating information.
2. Creating an API that allows users to submit reviews and ratings.
3. Designing a frontend UI that enables users to view and submit reviews and ratings.
4. Implementing filters so that users can sort dormitories based on review and rating categories.

3.6 Booking History

1. Incorporate a table in the database, for storing booking history information.
2. Build an API that enables users to access and review their dormitory booking history well as make new bookings.
3. Design a user frontend interface that allows users to conveniently view their dormitory bookings and make new ones.

3.7 Integration With School Calendar

1. Obtain the school's academic calendar data in a machine-readable format.
2. Develop a backend API to consume the school calendar data and store it in the database.
3. Update the frontend UI to display the school calendar data and ensure that students can book dormitories accordingly.

CHAPTER 4

LIMITATIONS OF THE STUDY

The proposed project is a complex system that handles dormitory operations. This broad scope causes one of the biggest project challenges as it can lead to delays, increased costs, and difficulties in project management. The project aims to develop an inclusive DMS that covers a wide range of features, including dormitory search, price comparison, service comparison, student pairing, rating and feedback, and dormitory owner interface.

4.1 Challenges

The project's grand scope presents several challenges. Firstly, the project also faces several time constraints. The proposal suggests that the project should be finished within a specific period. If development is delayed, it could negatively impact the project’s success, as students and dormitory owners may not be able to use the systems when they need it. Secondly, the development of a DMS is a costly undertaking. Thirdly, here are several risks associated with the DMS project, such as technical challenges, unforeseen delays, and user adoption. Technical challenges could arise during development, such as bugs and compatibility issues. Unforeseen delays could occur due to factors such as illness, staff turnover, and external events. User adoption is also a risk, as students and dormitory owners may not be willing to use the system if it is not user-friendly or does not provide value. Fourth, the quality of the DMS is another important consideration. Our system must be reliable, secure**,** and adaptable. If the system is not of high quality, it could experience performance issues, downtime, and security breaches. This could upset and annoy users and could damage the project's and the university's reputation. Additionally, it is vital to ensure that both all features are implemented correctly and work together seamlessly, yand that the system is scalable and can accommodate to many users without performance degradation. Finally, the benefits of the DMS project must also be carefully considered. The project proposal states that the system will simplify the process of finding suitable dormitory accommodation for students and will provide dormitory owners with a platform to manage their businesses more efficiently.

4.2 Mitigation Strategies

Many mitigation strategies can be used to address the limitations of the DMS project. These include:

- Scope management: The project scope should be carefully defined and managed.
- Time management: A realistic project plan should be developed and followed. This plan should include contingency plans to mitigate unforeseen delays.
- Cost management: A detailed budget should be developed and monitored closely. T This helps ensure that your project doesn't go over budget.
- Risk management: A risk management plan is needed to find, evaluate and minimize project risks.
- Quality Management: A quality control plan must be established to ensure the system meets standards.
- Benefits management: The benefits of the system should be carefully quantified and tracked. This ensures that the project provides added value for all involved.

By carefully considering the limitations of the DMS project and implementing appropriate mitigation strategies, the project team can increase the chances of success.

CHAPTER 5

LITERATURE REVIEW

Dormitory management involves allocating the rooms, safety and security, monitoring occupancy, and tracking student maintenance requests in the campus facilities. The primary problem of a dormitory management system is finding a solution that gives the ability to maintain tasks like room allocation, occupancy tracking, maintenance and repairs, security, communication, billing and payments, and data management.

As called before, allocating rooms to students considering their budget, preferences, and other factors is one of the most important tasks of a dormitory management system. Through communication with dormitory managers, it is estimated that preferring having roommates has increased each year in Northern Cyprus since 2020 so it is important to consider the applicator's preferences for roommate's characteristics.

The website also has to show which rooms are occupied, which are reserved but need a roommate, and which rooms are empty. It's one of the most important challenges, especially during peak admission times. Dorm owners and managers can check the status of rooms and the amount of full, reserved, or empty rooms. Also, the system has to show the duration of the contract that has been signed for each room. Students can also request maintenance and repairs for their rooms through this system so they can experience a comfortable residence in the dormitory. Ensuring security and safety through surveillance, control access, and reviews written by students is another task of this system. Students also can directly message the dormitory manager their complaints.

Facilitating communication between dormitory management and residents and also between residents through chat, status, comment, feedback, and review is also substantial. It is also important to manage and show the payments and debts accurately and transparently. Safety of information and protecting privacy are even more important tasks of the system. The dormitory management system should have the ability to record and manage data about the dormitory, residents, rooms,

payments, contracts, etc., and should keep the history of the dormitories, rooms, and residents and show the statistical analysis using the history.

5.1 What's already known

Several studies and research have shown the challenges of the dormitory management system and research and real-world application have shown the solution to these challenges. Advancing in information technology and its fields such as database management, cyber security, and IoT devices has helped a lot to achieve the goal of having safe and easy-to-use management systems.

One example of a system that uses information technology advantages in the real world is the University of Oregon's housing management website [1]. This website has used Hypertext Markup Language to build up its structure with the help of CSS to style this structure. As we go into more details, we can see that this website uses server-side scripting languages and databases as its backend development to reach goals like logging in or signing up on the website, exploring different buildings and room types available on the university campus, or connect to other students and managers. Technologies like PHP, Ruby on Rails, or Python can be used for server-side management of the website. Also, they may have used technologies like MySQL or NoSQL and databases like MongoDB to reach the goal of data storage and retrieval. It is also necessary to do UI design to ensure a consistent look and feel and UX design to optimize the website's accessibility and usability. The design of Oregan's website is also responsive, which means that this website is accessible and looks good on different devices such as laptop desktop, mobile, and tablet. The website needs to use web hosting services, a domain, and server configuration. A management system also needs best security practices to protect users' data from hacking and breaches. This can be easily achieved by using up-to-date software for servers, implementing strong password policies on the website, and enabling two-factor authentication, especially for dormitory administrators.

5.2 Previous Solutions in Dormitory Management

In general, many educational institutions around the world such as Brigham University in the USA [2] or the Federal Ministry of Education and Research in Germany [3] have used this idea to help students find their suitable housing around their universities or in their university's campus area. Northern Cyprus universities have done this also, but simpler. One of the examples is Eastern Mediterranean University located in Magusa which includes campus dormitories on its dedicated

website [4] including a description of the dormitory and its rooms, management contact information, many photos of rooms and floor plans, and bank accounts that students have to transfer money to in order to get a room. Other examples of universities in Northern Cyprus that use dormitory management websites are Middle East Technical University [5] and the University of Kyrenia [6].

5.3 Research and Innovations in Dormitory Systems

Many researches are out about dormitory management systems recommending the best way to implement these systems. One of them is Research on Dormitory Management System in Colleges and Universities by Lingfang Huang [7]. She and her team have found a student dormitory management system that is satisfied by its two clients, students and dormitory managers, through constant communication and surveys of university students and dormitory management personnel and data analysis. One of the main goals of this research is to obtain a user-friendly interface. In this article, it is mentioned that although many student dormitory management systems are available on the Internet and their functions are relatively complete, the most important issue is dormitory managers don't have the required knowledge to use the complicated interfaces and most of them are in an old age that learning new technologies is hard for them. So they stick to old-fashioned management using papers and face-to-face reservations. And although students are open to learning new technologies and knowledge, they also find these systems devastating to use. So, ensuring that the functionality of a website is complete is not enough. The system should be user-friendly and convenient to use. Huang has found the best design for the development of student dormitory management through feasibility analysis of system development, users' needs analysis, testing, and improvement of the system that the requirements were gathered from surveys and communications. More information including the flow diagram of system data, structural design of system concept, physical design, and system implementation and main codes are available in her article.

Another significant article including dormitory management systems is Automated Dormitory Management System by Tajebe Tsega [8] from Bule Hor University. The purpose of his project was to develop computerized dormitory management for his university and he has included the problems this university has faced before when they wanted to shift the dormitory management from manual to automated, the process of manual dormitory management, why there should be an online management system for this issue, his proposed web-based system for Bule Hor university,

and the benefits and scope of this system. He also wrote about different methodologies of his project, including data collection, analysis, and design methods in his paper. One interesting thing he has done in this article is finding and putting the costs of the project in a tabular format and comparing it with the manual process of dormitory management on the university campus. This article also includes different diagrams such as sequence diagrams of different functionalities of the proposed system, different designs of this system such as a three-tier architectural design, and many tables for required data. All these pieces of information make this article remarkable and usable.

CHAPTER 6

METHODOLOGY OF THE PROPOSED STUDY

The proposed methodology for the Dormitory Management System (DMS) project is an agile development approach. Agile development is a way of building software that focuses on delivering working features to users quickly and often. The agile development methodology is a good fit for the DMS project because it allows the project team to release working features to users quickly and get feedback early on. The agile development process is an iterative and incremental approach to software development that focuses on delivering working software early and often. Phases of the proposed methodology:

- **Planning**. In this phase the team defines the project scope, all features, and functionalities.
- **Development**: During the development phase, the project team implements the features and functionality defined in the planning phase. During the development phase, the project team implements the features and functionality defined in the planning phase. This will be done in a series of sprints, which are short, time-boxed periods of work.
- The third phase of the agile development process is **testing**. In testing, the team makes sure that everything works with no errors and bugs.
- The **deployment** phase rolls out the system to production and makes it available to users.

The project team will use the following tools to discuss the tasks and track the progress:

- **Telegram** - for communication and task discussions.
- **GitHub** - for hosting code and collaboration.

The project team will also use a variety of reporting tools to track progress and identify potential risks. These tools may include:

- **Burn-down charts:** To track the progress of each sprint, the project team will use burn-down charts.
- **Velocity charts:** To track the team's performance over time, the project team will use velocity charts.

- **Risk registers:** To identify and track risks, the project team will use a risk register.

Problem Solution

The proposed DMS system will solve the following problems:

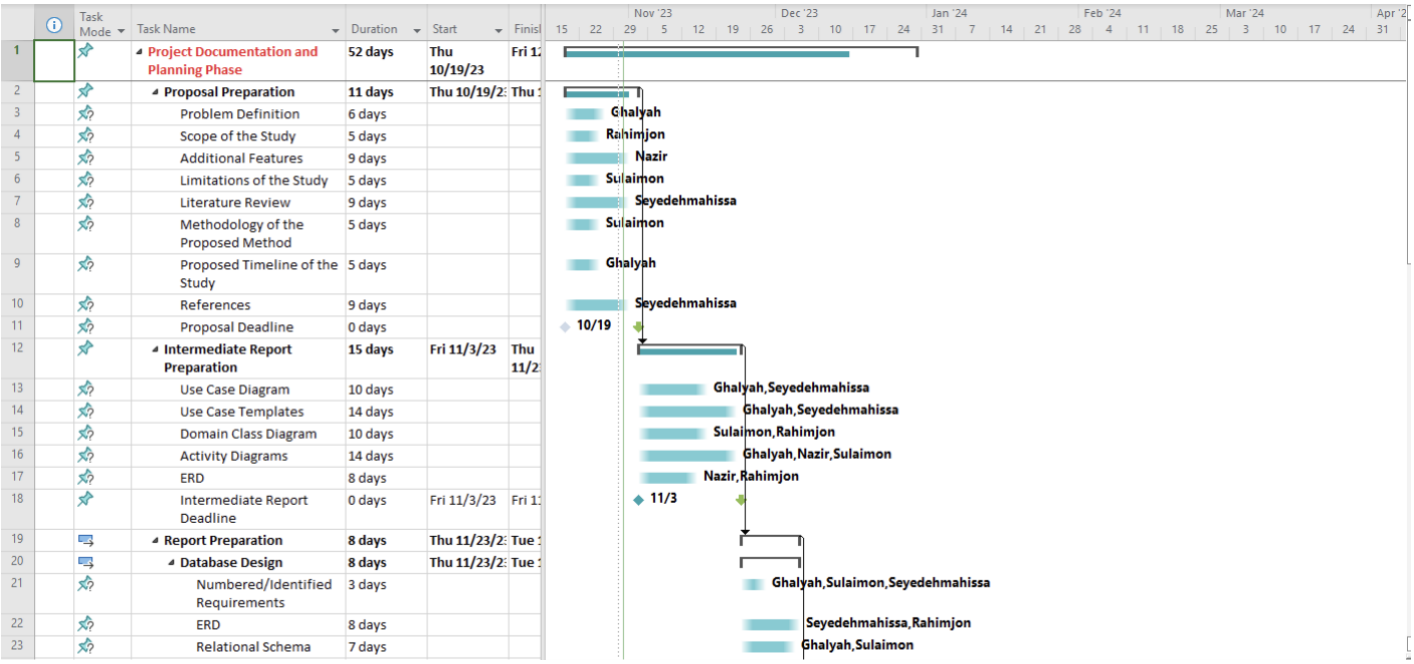
- It will be difficult for students to find suitable dormitory accommodation.
- It will be difficult for dormitory owners to manage their businesses.
- The university will not have a centralized system to track dormitory occupancy and other data.

The DMS system will provide the following solutions:

- Students will be able to search for and book dormitories online.
- Dormitory owners will be able to manage their listings and reservations online.
- The university will have access to data on dormitory occupancy and other metrics.

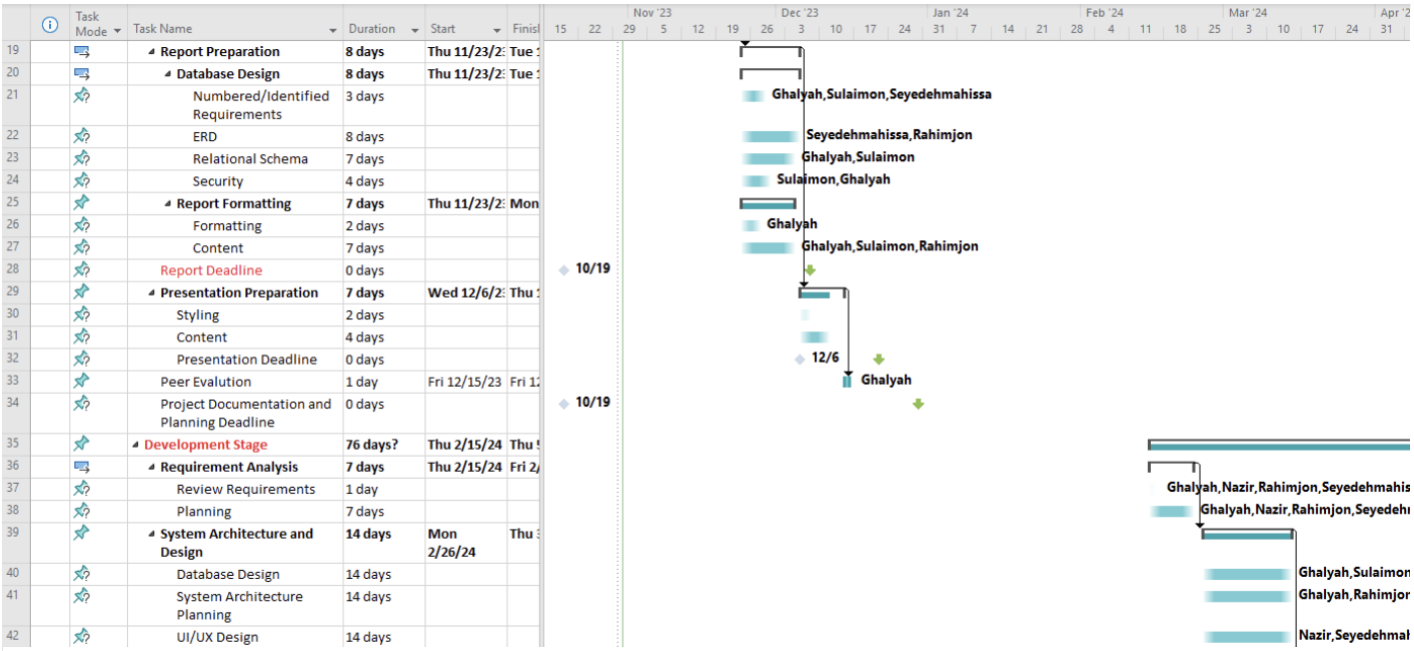
CHAPTER 7

PROPOSED TIMELINE OF THE STUDY



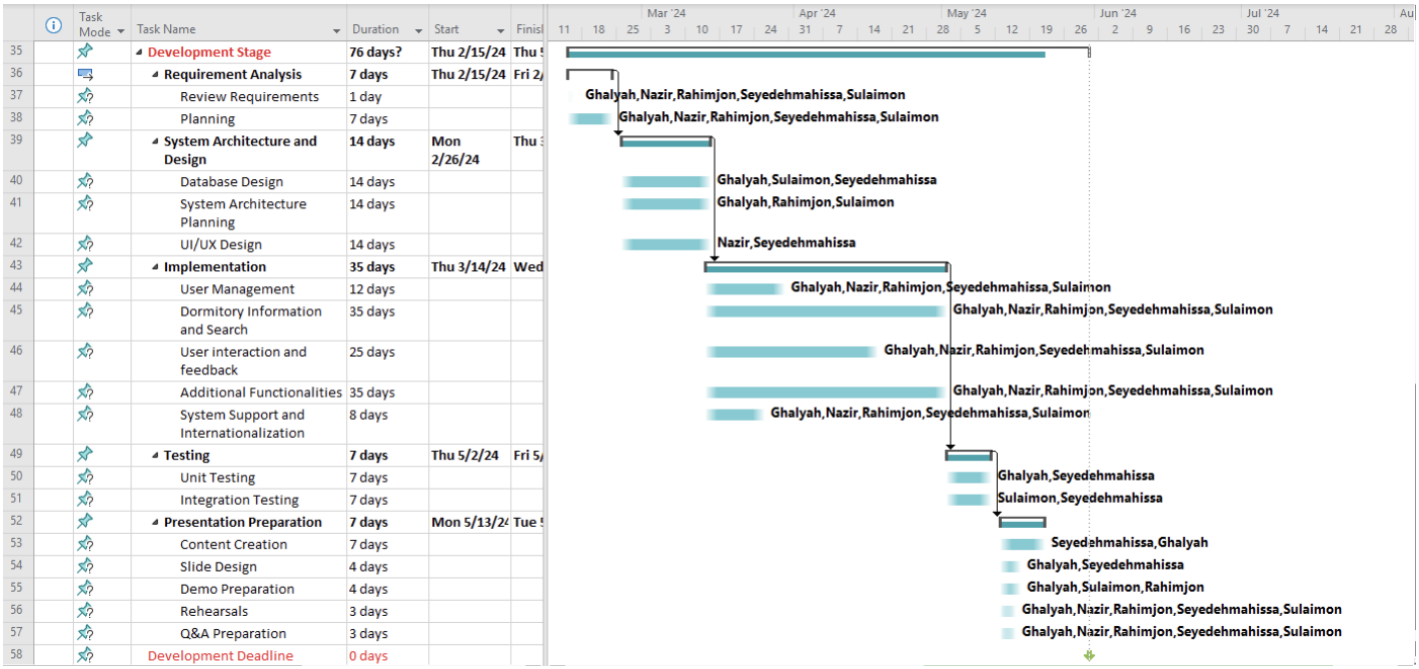
	Task Name	Duration	Start	Finish
1	Project Documentation and Planning Phase	52 days	Thu 10/19/23	Fri 1
2	Proposal Preparation	11 days	Thu 10/19/2	Thu
3	Problem Definition	6 days		
4	Scope of the Study	5 days		
5	Additional Features	9 days		
6	Limitations of the Study	5 days		
7	Literature Review	9 days		
8	Methodology of the Proposed Method	5 days		
9	Proposed Timeline of the Study	5 days		
10	References	9 days		
11	Proposal Deadline	0 days		
12	Intermediate Report Preparation	15 days	Fri 11/3/23	Thu 11/2
13	Use Case Diagram	10 days		
14	Use Case Templates	14 days		
15	Domain Class Diagram	10 days		
16	Activity Diagrams	14 days		
17	ERD	8 days		
18	Intermediate Report Deadline	0 days	Fri 11/3/23	Fri 1
19	Report Preparation	8 days	Thu 11/23/2	Tue
20	Database Design	8 days	Thu 11/23/2	Tue
21	Numbered/Identified Requirements	3 days		
22	ERD	8 days		
23	Relational Schema	7 days		

Figure 7.2: Project Documentation and Planning Stage



	Task Name	Duration	Start	Finish
19	Report Preparation	8 days	Thu 11/23/2	Tue
20	Database Design	8 days	Thu 11/23/2	Tue
21	Numbered/Identified Requirements	3 days		
22	ERD	8 days		
23	Relational Schema	7 days		
24	Security	4 days		
25	Report Formatting	7 days	Thu 11/23/2	Mon
26	Formatting	2 days		
27	Content	7 days		
28	Report Deadline	0 days		
29	Presentation Preparation	7 days	Wed 12/6/2	Thu
30	Styling	2 days		
31	Content	4 days		
32	Presentation Deadline	0 days		
33	Peer Evalution	1 day	Fri 12/15/23	Fri 1
34	Project Documentation and Planning Deadline	0 days		
35	Development Stage	76 days?	Thu 2/15/24	Thu
36	Requirement Analysis	7 days	Thu 2/15/24	Fri 2
37	Review Requirements	1 day		
38	Planning	7 days		
39	System Architecture and Design	14 days	Mon 2/26/24	Thu
40	Database Design	14 days		
41	System Architecture Planning	14 days		
42	UI/UX Design	14 days		

Figure 7.1: Project Documentation and Planning Stage



	Task Mode	Task Name	Duration	Start	Finish
35		Development Stage	76 days?	Thu 2/15/24	Thu 5
36		Requirement Analysis	7 days	Thu 2/15/24	Fri 2/
37		Review Requirements	1 day		
38		Planning	7 days		
39		System Architecture and Design	14 days	Mon 2/26/24	Thu 3
40		Database Design	14 days		
41		System Architecture Planning	14 days		
42		UI/UX Design	14 days		
43		Implementation	35 days	Thu 3/14/24	Wed
44		User Management	12 days		
45		Dormitory Information and Search	35 days		
46		User interaction and feedback	25 days		
47		Additional Functionalities	35 days		
48		System Support and Internationalization	8 days		
49		Testing	7 days	Thu 5/2/24	Fri 5/
50		Unit Testing	7 days		
51		Integration Testing	7 days		
52		Presentation Preparation	7 days	Mon 5/13/24	Tue 5
53		Content Creation	7 days		
54		Slide Design	4 days		
55		Demo Preparation	4 days		
56		Rehearsals	3 days		
57		Q&A Preparation	3 days		
58		Development Deadline	0 days		

Figure 7.3: Development Stage

CHAPTER 8

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